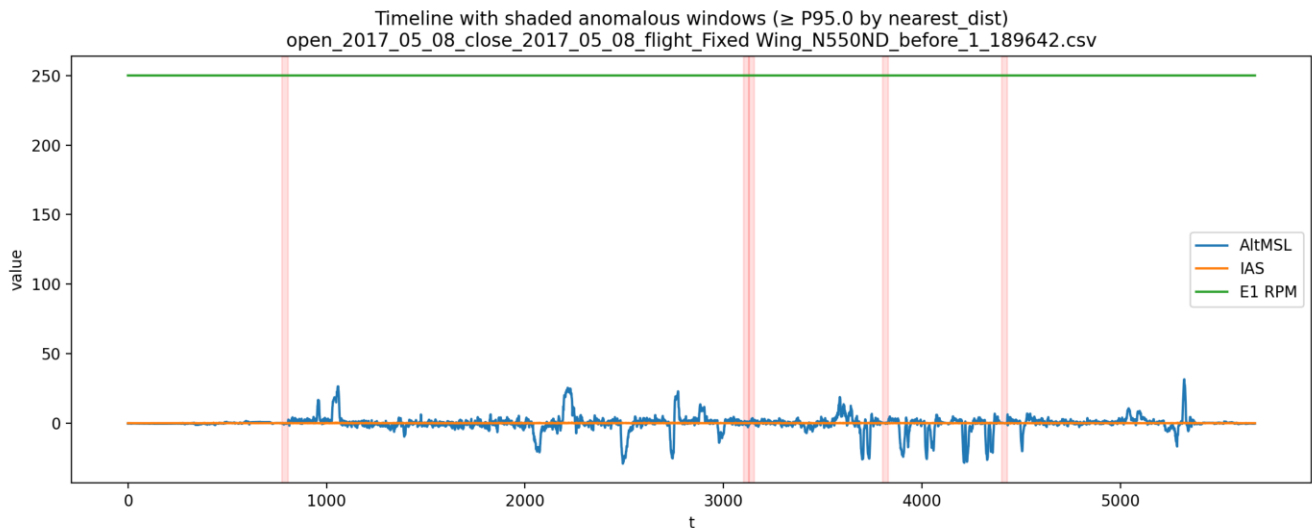


NGAFID Anomaly Detection — One-Pager

We train an OC-SVM on healthy flight windows and learn prototype “boxes” from K-Means. During inference, each window’s distance to the nearest prototype and per-sensor box violations flag anomalies and explain which channels deviated (e.g., AltMSL, IAS, RPM).

Example timeline with shaded anomalies ($\geq$ P95 nearest_dist)



Calibration (RAW-based thresholds applied to ERROR)

signal	threshold	thr_value	precision	recall	f1
nearest_dist	P95	14.81	0.938	0.774	0.848
nearest_dist	P99	18.49	0.984	0.672	0.799
viol_count_sum	P95	73.0	0.949	0.97	0.959
viol_count_sum	P99	105.89	0.988	0.907	0.946
viol_sev_sum	P95	367.0	0.949	0.946	0.947
viol_sev_sum	P99	558.14	0.988	0.918	0.952